# OYEN AGDM, AB, Canada

WMO: **713420**

Lat: **51.3797N** Lon: **110.3552W** Elev: **767** StdP: **92.45** Time Zone: **-7.00 (NAM)** Period: **02-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **2** | **-29.5** | **-26.7** | **-33.1** | **0.2** | **-29.4** | **-30.1** | **0.3** | **-26.7** | **13.2** | **-4.8** | **11.7** | **-5.6** | **3.7** | **250** | **0.736** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **13.7** | **31.1** | **17.0** | **28.9** | **16.3** | **26.8** | **15.8** | **19.1** | **26.7** | **17.9** | **25.3** | **16.9** | **24.2** | **4.7** | **190** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **16.7** | **13.1** | **21.6** | **15.3** | **12.0** | **20.6** | **14.2** | **11.1** | **19.2** | **57.5** | **26.7** | **53.7** | **25.4** | **50.4** | **24.4** | **24.7** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **12.3** | **10.7** | **9.4** | DB | **-33.5** | **34.3** | **2.6** | **2.0** | **-35.3** | **35.8** | **-36.8** | **37.0** | **-38.3** | **38.1** | **-40.2** | **39.6** | (4) |
| (5) | | | | WB | **-33.3** | **21.2** | **2.7** | **1.6** | **-35.3** | **22.3** | **-36.9** | **23.3** | **-38.4** | **24.1** | **-40.4** | **25.3** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | DBAvg | | **3.8** | **-10.2** | **-11.0** | **-3.3** | **4.3** | **10.9** | **15.1** | **18.4** | **17.1** | **12.1** | **4.8** | **-3.3** | **-10.0** | (6) |
| (7) | | DBStd | | **12.18** | **8.25** | **7.78** | **7.91** | **5.58** | **4.59** | **3.40** | **3.26** | **3.63** | **4.90** | **5.56** | **6.90** | **7.92** | (7) |
| (8) | | HDD10.0 | | **3088** | **626** | **588** | **414** | **180** | **46** | **3** | **0** | **1** | **32** | **177** | **401** | **620** | (8) |
| (9) | | HDD18.3 | | **5390** | **884** | **821** | **672** | **421** | **234** | **108** | **40** | **68** | **194** | **420** | **650** | **878** | (9) |
| (10) | | CDD10.0 | | **828** | **0** | **0** | **1** | **9** | **73** | **154** | **259** | **221** | **96** | **14** | **1** | **0** | (10) |
| (11) | | CDD18.3 | | **89** | **0** | **0** | **0** | **0** | **2** | **11** | **41** | **29** | **7** | **0** | **0** | **0** | (11) |
| (12) | | CDH23.3 | | **1457** | **0** | **0** | **1** | **3** | **73** | **170** | **580** | **462** | **162** | **6** | **0** | **0** | (12) |
| (13) | | CDH26.7 | | **474** | **0** | **0** | **0** | **0** | **12** | **42** | **208** | **157** | **55** | **0** | **0** | **0** | (13) |
| (14) | Wind | WSAvg | | **4.8** | **5.3** | **4.8** | **5.1** | **5.3** | **5.2** | **4.8** | **4.2** | **4.1** | **4.4** | **5.0** | **4.9** | **4.8** | (14) |
| (15) | Precipitation | PrecAvg | | **330** | **14** | **8** | **13** | **21** | **37** | **73** | **53** | **43** | **25** | **14** | **13** | **11** | (15) |
| (16) | | PrecMax | | **471** | **47** | **18** | **36** | **61** | **80** | **150** | **128** | **84** | **83** | **33** | **44** | **18** | (16) |
| (17) | | PrecMin | | **172** | **2** | **1** | **3** | **3** | **3** | **17** | **20** | **5** | **1** | **4** | **0** | **2** | (17) |
| (18) | | PrecStd | | **75** | **10** | **5** | **7** | **15** | **23** | **32** | **24** | **21** | **21** | **7** | **11** | **5** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **6.0** | **5.2** | **15.8** | **23.2** | **28.4** | **30.9** | **34.4** | **33.2** | **31.7** | **24.1** | **15.2** | **6.9** | (19) |
| (20) | | | MCWB | | **2.6** | **2.3** | **8.1** | **10.7** | **14.1** | **17.3** | **18.5** | **16.5** | **15.4** | **12.6** | **7.9** | **3.5** | (20) |
| (21) | | 2% | DB | | **4.1** | **3.1** | **11.9** | **19.5** | **25.4** | **27.5** | **31.3** | **30.5** | **27.8** | **19.7** | **10.2** | **4.5** | (21) |
| (22) | | | MCWB | | **1.7** | **0.9** | **5.9** | **9.2** | **12.6** | **15.5** | **18.3** | **16.2** | **14.3** | **10.4** | **5.1** | **2.0** | (22) |
| (23) | | 5% | DB | | **2.7** | **1.5** | **8.9** | **16.4** | **22.9** | **25.1** | **29.0** | **28.3** | **24.5** | **16.1** | **7.4** | **2.8** | (23) |
| (24) | | | MCWB | | **0.8** | **-0.2** | **4.4** | **7.5** | **11.7** | **15.2** | **17.2** | **16.1** | **13.3** | **8.6** | **3.4** | **0.6** | (24) |
| (25) | | 10% | DB | | **1.0** | **-0.4** | **6.0** | **13.7** | **20.4** | **22.8** | **26.6** | **25.8** | **21.4** | **13.2** | **4.8** | **0.4** | (25) |
| (26) | | | MCWB | | **-0.6** | **-1.7** | **2.7** | **6.3** | **10.5** | **14.2** | **17.0** | **15.5** | **12.0** | **7.2** | **1.9** | **-1.2** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **3.0** | **2.7** | **8.6** | **11.4** | **15.1** | **19.0** | **21.8** | **19.4** | **16.3** | **13.2** | **8.2** | **3.9** | (27) |
| (28) | | | MCDB | | **5.3** | **4.8** | **15.1** | **21.6** | **24.8** | **27.4** | **28.9** | **27.0** | **27.9** | **22.4** | **14.8** | **6.3** | (28) |
| (29) | | 2% | WB | | **1.8** | **1.1** | **6.2** | **9.6** | **13.6** | **17.2** | **20.0** | **18.1** | **14.9** | **10.9** | **5.6** | **2.2** | (29) |
| (30) | | | MCDB | | **3.7** | **2.8** | **11.0** | **18.1** | **23.1** | **24.3** | **27.3** | **26.3** | **25.9** | **18.2** | **9.6** | **4.4** | (30) |
| (31) | | 5% | WB | | **0.9** | **-0.1** | **4.5** | **8.1** | **12.4** | **16.0** | **18.7** | **17.0** | **13.8** | **9.2** | **3.7** | **0.6** | (31) |
| (32) | | | MCDB | | **2.7** | **1.5** | **8.5** | **15.3** | **21.0** | **23.2** | **26.1** | **25.1** | **23.3** | **15.2** | **6.7** | **2.6** | (32) |
| (33) | | 10% | WB | | **-0.6** | **-1.7** | **2.7** | **6.8** | **11.3** | **14.9** | **17.7** | **16.1** | **12.5** | **7.6** | **2.1** | **-1.1** | (33) |
| (34) | | | MCDB | | **1.0** | **-0.3** | **5.7** | **12.6** | **19.0** | **21.2** | **24.6** | **23.8** | **20.4** | **12.8** | **4.5** | **0.4** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **9.5** | **9.4** | **10.2** | **12.1** | **13.2** | **11.8** | **13.7** | **13.7** | **13.2** | **11.7** | **9.2** | **8.7** | (35) |
| (36) | | 5% DB | MCDBR | | **9.3** | **10.2** | **13.7** | **17.2** | **17.1** | **15.7** | **17.7** | **18.4** | **18.5** | **16.3** | **12.1** | **9.7** | (36) |
| (37) | | | MCWBR | | **7.6** | **8.3** | **8.4** | **8.7** | **7.7** | **6.8** | **7.1** | **7.1** | **7.8** | **8.6** | **7.9** | **7.7** | (37) |
| (38) | | 5% WB | MCDBR | | **9.2** | **10.0** | **12.9** | **15.6** | **14.9** | **13.7** | **15.1** | **15.5** | **17.2** | **15.5** | **11.3** | **9.5** | (38) |
| (39) | | | MCWBR | | **7.7** | **8.4** | **8.3** | **8.2** | **7.1** | **6.7** | **7.2** | **6.7** | **7.6** | **8.5** | **7.8** | **7.6** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.260** | **0.293** | **0.295** | **0.348** | **0.375** | **0.367** | **0.368** | **0.383** | **0.329** | **0.289** | **0.262** | **0.252** | (40) |
| (41) | | taud | | | **2.354** | **2.308** | **2.436** | **2.301** | **2.266** | **2.348** | **2.376** | **2.296** | **2.434** | **2.519** | **2.473** | **2.363** | (41) |
| (42) | | Ebn at Noon | | | **774** | **834** | **913** | **890** | **877** | **887** | **880** | **843** | **861** | **842** | **779** | **725** | (42) |
| (43) | | Edh at Noon | | | **60** | **84** | **90** | **117** | **127** | **118** | **113** | **116** | **90** | **68** | **54** | **51** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **1.17** | **2.23** | **3.59** | **4.86** | **5.85** | **6.08** | **6.54** | **5.32** | **3.82** | **2.32** | **1.23** | **0.88** | (44) |
| (45) | | RadStd | | | **0.12** | **0.20** | **0.27** | **0.39** | **0.39** | **0.38** | **0.28** | **0.39** | **0.34** | **0.28** | **0.11** | **0.09** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (49 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page